

Predicting Corporate Default: A Dynamic Logit Benchmark versus Machine Learning

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Abstract

We ask whether a flexible machine-learning model improves corporate default prediction relative to a standard dynamic logit benchmark. Using a firm-year panel of U.S. non-financial, non-utility public firms built from Compustat, CRSP, and CRSP delisting files, we predict default from accounting ratios, market-valued balance-sheet measures, stock-price information, and liquidity variables. Default events are proxied by CRSP distress-related delisting codes, and all predictors are measured strictly before the default window, so the design is point-in-time. We compare a dynamic logit model, motivated by the predictor set in [Campbell et al. \(2008\)](#), with an XGBoost classifier trained on the same panel, features, and outcome. Both are trained on pre-2006 observations and evaluated out of sample over three later windows: 2006–2010, 2011–2015, and post-2015. Both discriminate well, with AUC-ROC values around 0.90–0.93. XGBoost achieves higher AUC-ROC, precision-recall AUC, and KS in all three windows, but the AUC-ROC gain is significant only in 2011–2015 (DeLong test, $p = 0.001$), the one window containing no macro-shock year and so the cleanest read on the comparison; the most consistent improvement is in precision-recall AUC, the relevant metric when defaults are rare. The gains are meaningful but not transformative: the dynamic logit remains competitive, transparent, and economically interpretable. Out-of-sample pseudo- R^2 , measured against a forecast null anchored to the training base rate, also favors XGBoost, though the models are nearly tied in 2006–2010. Calibration behaves differently from ranking. Splitting each window into stress and ordinary years shows that both models track realized default rates closely outside two macro-shock episodes and fail in opposite directions inside them: understating defaults by about half in 2007–2008 and overstating them by roughly an order of magnitude in 2020–2022. Window-level ratios average these episodes with well-calibrated ordinary years, making a concentrated, regime-specific failure look like gradual drift. The central finding is a decoupling: discrimination is stable across every window we test, while probability levels hold only under the conditions the models were fitted for. Flexible models can therefore improve default screening, but expected-loss applications require probabilities validated against the macro environment in which they are used.

Keywords: corporate default; bankruptcy prediction; dynamic logit; discrete-time hazard model; XGBoost; credit risk; probability calibration.

1 Introduction

Predicting which firms will experience financial distress is central to credit risk measurement. Default forecasts affect debt pricing, portfolio monitoring, expected-loss calculations, and capital planning. A large literature has built default prediction models from accounting ratios, beginning with [Altman \(1968\)](#) and [Ohlson \(1980\)](#), from discrete-time hazard models that use the full firm-year panel structure, as in [Shumway \(2001\)](#), and from market-based predictors that reprice firm distress in real time, as in [Campbell et al. \(2008\)](#). In parallel, machine-learning methods, including tree ensembles, have become standard tools for prediction problems with nonlinearities and interactions. The practical question for this project is whether that flexibility improves corporate default prediction relative to a strong and interpretable dynamic logit benchmark.

This paper answers that question directly. We are *not* attempting to replicate or extend [Campbell et al. \(2008\)](#) exactly. Rather, we borrow the main predictor logic from that framework: scaling profitability, leverage, and liquidity by the market value of assets, and combining these measures with stock-price information and standard accounting ratios. We use these variables to construct a strong traditional benchmark and then compare that benchmark head to head with a machine-learning model estimated on the same panel, the same selected predictors, and the same outcome.

Our traditional model is a dynamic logit model, interpreted as a discrete-time hazard model, in the tradition of [Shumway \(2001\)](#) and [Campbell et al. \(2008\)](#). The model is estimated on a firm-year panel and remains linear in the log-odds, transparent, and economically interpretable. Our machine-learning model is XGBoost, a gradient-boosted tree ensemble. It is trained on the same firm-year panel and predictor set, with hyperparameters selected by a time-series split within the training sample. Because predicted probabilities from boosted trees are not automatically well calibrated, we also apply sigmoid calibration, following the logic of Platt scaling.

We compare the two models along four dimensions: *discrimination*, measured by AUC-ROC, precision-recall AUC, and the Kolmogorov–Smirnov statistic; *goodness of fit*, measured by in-sample and out-of-sample pseudo- R^2 ; *calibration*, measured by comparing average predicted default probabilities with realized default rates; and *expected-loss implications*, illustrated using predicted default probabilities, debt exposure, and an assumed loss-given-default. All evaluation is out of time: the models are trained on pre-2006 data and tested on three later, non-overlapping windows. In addition, all predictors are measured at fiscal year-end and lagged so that they strictly precede the default window, giving a point-in-time design with no look-ahead from the feature side.

Three findings stand out. First, both models discriminate well out of sample, with AUC-ROC values between 0.90 and 0.93. Second, the machine-learning model achieves higher AUC-ROC, precision-recall AUC, and KS than the dynamic logit in every test window, but the AUC-ROC improvement is statistically significant only in 2011–2015 (DeLong test, $p = 0.001$); the most consistent gains are in precision-recall AUC, the most informative metric when defaults are rare. Goodness-of-fit results, measured by out-of-sample pseudo- R^2 against a forecast null anchored to the training base rate, also favor XGBoost in all three windows, though only marginally in the

2006–2010 window.

Third, probability calibration is much less stable than ranking, and the instability is concentrated in particular years rather than spread across the later sample. Splitting each test window into stress and ordinary years shows that both models track realized default rates closely in ordinary years and fail only during two macro-shock episodes. In 2007–2008 both underpredict by roughly half, with ratios of average predicted probability to realized default rate of 0.51 for the dynamic logit and 0.56 for XGBoost, while the ordinary years of the 2006–2010 window calibrate well (0.96 and 1.13). The direction reverses in 2020–2022, when predicted probabilities exceed realized distress-delisting rates by more than an order of magnitude; that ratio rests on a handful of realized defaults, so we read it as directionally real rather than as a precise estimate. The 2016–2019 years that sit between the two episodes are again close to right (1.07 and 1.15). Window-level calibration averages are therefore blends of a well-calibrated ordinary-year subsample and a badly miscalibrated stress-year subsample, not a uniform property of any window.

This also gives 2011–2015 a specific role in the comparison. It is the only test window containing no stress year, which makes it the cleanest read on how the two models differ, and it is the window in which the AUC-ROC gap is significant. The two results should not be collapsed into one, however: dropping stress years from the other two windows leaves the AUC-ROC comparison statistically inconclusive there as well, so the modest discrimination edge is not an artifact of stress-year noise. The central pattern is a decoupling of ranking from calibration. XGBoost sharpens the ordering of firms by risk, while for both models the probability levels that feed expected-loss calculations hold up in ordinary years and break down, in both directions, during named macro shocks.

The paper proceeds as follows. Section 2 describes the data sources, sample construction, default definition, predictor variables, and descriptive statistics. Section 3 presents the dynamic logit and XGBoost models, along with probability calibration, evaluation metrics, and the out-of-time design. Section 4 reports the empirical results, including the dynamic-logit benchmark, the XGBoost model, and the direct model comparison. Section 5 discusses calibration around macro shocks and expected-loss implications. Section 6 concludes.

2 Data

2.1 Sources and sample construction

We construct a firm-year panel by merging three WRDS data sources. First, we use Compustat annual fundamentals for accounting variables. Second, we use CRSP monthly stock data for prices, returns, shares outstanding, and market-based predictors. Third, we use CRSP delisting records to identify distress-related exits from public markets. Compustat firms are linked to CRSP securities using the CRSP/Compustat Merged link identifier, `lpermno`, and are matched to CRSP monthly observations using the fiscal year-end month. This matching procedure allows firms with non-December fiscal years to be aligned with the appropriate stock-market information.

Following standard practice, we exclude financial firms, with SIC codes 6000–6999, and regulated

utilities, with SIC codes 4900–4999. These industries have accounting structures and regulatory environments that differ substantially from industrial firms. For example, leverage has a different interpretation for banks, and utilities operate under rate regulation. Including these sectors would therefore mix firms with very different balance-sheet structures and default mechanisms. All continuous features are winsorized annually at the 1st and 99th percentiles of the cross-sectional distribution; computing the bounds within each fiscal year reduces the influence of data errors and extreme outliers while ensuring that no future information enters the transformation.

After merging the accounting, market, and delisting data, we remove duplicate firm-year observations. When a Compustat firm links to more than one CRSP security in the same fiscal year, we keep the security with the highest market capitalization, which is intended to capture the primary share class. We also remove firms whose Compustat observations appear after their recorded delisting year, since their event timing is unreliable. The final analysis panel, after requiring all selected model features to be non-missing, contains 157,556 firm-year observations for 16,928 unique firms. The sample period is 1980–2022.

2.2 Default definition

The outcome variable is a default-like distress event constructed from CRSP delisting records. We define a default event when the delisting reason type corresponds to bankruptcy, insolvency, or delinquency, using the CRSP codes `BKPY`, `INSC`, and `DELQ`. This definition is broader than formal bankruptcy alone: it is intended to capture public firms that exit the exchange under financial distress even if they do not appear as Chapter 11 bankruptcies.

For each firm with a distress-related delisting record, the default indicator is assigned to its last complete fiscal year in the cleaned Compustat–CRSP panel, and we retain a defaulting firm only when this final fiscal year precedes the delisting year by one or two years. All predictors are measured at that fiscal year-end, so they are observed strictly before the delisting event, and the model predicts default over a one- to two-year forward horizon rather than contemporaneously. Each defaulting firm therefore contributes at most one default event, and this timing convention ensures that no information from the delisting year or later enters the predictor set. The outcome should still be interpreted as a firm-year distress measure rather than an exact filing-date bankruptcy measure.

Across the final panel there are 1,870 default events, corresponding to an overall event rate of 1.187%. The realized default rate declines over time, from 1.404% in the pre-2006 training window to 0.481% after 2015. This declining base rate matters for interpreting precision-recall performance, since the no-skill AUC-PR baseline is the realized default prevalence in each window. Detailed event counts and rates by period are reported in Appendix Table A1.

2.3 Feature construction

The cleaning pipeline constructs a broader set of accounting and market variables, but the final dynamic logit and XGBoost specifications use the same **13 selected predictors**, listed in Table 1.

We group them into three categories. The first group consists of six book-based accounting variables: book leverage, interest coverage, ROA, current ratio, cash/assets, and log assets. These variables capture debt burden, debt-service capacity, profitability, liquidity, and firm size.

The second group consists of market-valued measures motivated by [Campbell et al. \(2008\)](#). These variables scale profitability, leverage, and liquidity by a market-based measure of firm value rather than by book assets. We define market-valued assets as

$$\text{MVA} = \text{market equity} + \text{total liabilities}.$$

This scaling is useful because market equity adjusts quickly when investors revise their beliefs about firm value. A firm whose equity value has already collapsed will therefore appear more distressed in these measures even before the next annual accounting statement is released.

The main market-valued variables are

$$\text{NIMTA} = \frac{\text{net income}}{\text{market equity} + \text{total liabilities}}, \quad \text{market-scaled profitability}, \quad (1)$$

$$\text{TLMTA} = \frac{\text{total liabilities}}{\text{market equity} + \text{total liabilities}}, \quad \text{market-scaled leverage}, \quad (2)$$

$$\text{CASHMTA} = \frac{\text{cash and short-term investments}}{\text{market equity} + \text{total liabilities}}, \quad \text{market-scaled liquidity}, \quad (3)$$

$$\text{Log price} = \min\{\log(|\text{price}| + 1), \log(15)\}, \quad \text{log stock price, capped at \$15}. \quad (4)$$

We cap log price at $\log(15)$ following [Campbell et al. \(2008\)](#)'s treatment of low-priced stocks: penny-stock prices are noisy and driven by microstructure effects, so the cap limits their influence on the fit without discarding the observations.

The third group contains additional market variables: the trailing 12-month equity return, annualized equity volatility, and the market-to-book ratio. These variables capture recent stock performance, market uncertainty, and valuation. Following [Campbell et al. \(2008\)](#), we include the components of distance-to-default—equity volatility, market leverage, and log price—as separate predictors rather than combining them into a single Merton distance-to-default index. This allows the model to weight each component freely rather than imposing the structural relationships implied by the [Merton \(1974\)](#) option-pricing framework; see also [Bharath and Shumway \(2008\)](#).

2.4 Descriptive statistics

Before estimating the models, we examine the distribution of the predictor variables used in both the dynamic logit and XGBoost specifications. Table 2 reports the number of observations and summary statistics for all 13 selected model features across the training and test windows. Each feature cell reports the mean, with the standard deviation in parentheses. The variables differ substantially in scale because the feature set combines accounting ratios, market-valued balance-sheet measures, stock returns, volatility, valuation ratios, and firm size. All features are winsorized annually as

Table 1: Feature definitions

Feature	Definition
<i>Panel A: Accounting variables</i>	
Book leverage	(long-term debt + short-term debt)/total assets
Interest coverage	EBITDA/interest expense; set to a high capped value (maximal coverage) when interest expense is zero
ROA	net income/total assets
Current ratio	current assets/current liabilities
Cash/assets	cash and short-term investments/total assets
Log assets	$\log(\text{total assets})$
<i>Panel B: Market-valued variables</i>	
NIMTA	net income/(market equity + total liabilities)
TLMTA	total liabilities/(market equity + total liabilities)
CASHMTA	cash and short-term investments/(market equity + total liabilities)
Log price	$\min\{\log(\text{price} + 1), \log(15)\}$
<i>Panel C: Other market variables</i>	
Trailing 12-month return	Compounded 12-month equity return ending in the fiscal year-end month
Equity volatility	Annualized standard deviation of monthly returns over the trailing 12 months
Market-to-book	market capitalization/book equity

Notes. Both models use all 13 selected features in Panels A–C. Continuous features are winsorized annually at the 1st and 99th percentiles of the cross-sectional distribution. Because a firm with no interest expense has no debt-servicing burden, interest coverage is assigned a high capped value in that case (maximal coverage) rather than zero, which would incorrectly mark such firms as the least covered.

described in Section 2.1.

The table also shows that the distribution of firm characteristics changes across test periods. Book leverage rises in the post-2015 window, for example, and the dispersion of interest coverage and market-to-book widens. Because the models are trained once on pre-2006 data and never refit, these shifts in the predictor distribution are part of what the out-of-sample exercise is testing.

Table 2: Summary statistics for selected model features

Feature	Train	2006–2010	2011–2015	Post-2015
N	103,154	17,298	15,475	21,629
Book leverage	0.238 (0.209)	0.203 (0.215)	0.223 (0.219)	0.287 (0.246)
Interest coverage	26.614 (118.423)	54.629 (219.947)	49.043 (226.417)	3.244 (259.204)
ROA	-0.051 (0.288)	-0.046 (0.280)	-0.037 (0.256)	-0.095 (0.359)
Current ratio	2.784 (2.917)	2.831 (2.697)	2.871 (2.914)	2.939 (3.430)
Cash/assets	0.160 (0.196)	0.219 (0.227)	0.212 (0.222)	0.237 (0.251)
Log assets	4.823 (2.161)	6.206 (2.092)	6.615 (2.116)	6.833 (2.199)
NIMTA	-0.024 (0.172)	-0.026 (0.179)	-0.018 (0.147)	-0.058 (0.224)
TLMTA	0.397 (0.258)	0.368 (0.251)	0.372 (0.248)	0.385 (0.256)
CASHMTA	0.106 (0.150)	0.146 (0.182)	0.137 (0.170)	0.149 (0.218)
Log price	2.067 (0.755)	2.170 (0.717)	2.230 (0.692)	2.215 (0.722)
Trailing 12-month return	0.155 (0.733)	0.126 (0.709)	0.082 (0.501)	0.077 (0.644)
Equity volatility	0.536 (0.335)	0.504 (0.301)	0.413 (0.235)	0.522 (0.355)
Market-to-book	2.571 (3.939)	2.488 (3.829)	2.772 (5.401)	3.393 (8.240)

Notes. Each cell reports the mean with the standard deviation in parentheses, except for the first row, which reports the number of firm-year observations. Train includes firm-year observations through 2005. The three test windows cover 2006–2010, 2011–2015, and post-2015 observations. All features are winsorized annually at the 1st and 99th percentiles of the cross-sectional distribution.

3 Methodology

3.1 Dynamic logit

The traditional benchmark is a dynamic logit model, interpreted as a discrete-time hazard model, following the logic of Shumway (2001) and Campbell et al. (2008). Let $Y_{it} = 1$ if firm i defaults—that is, experiences a distress-related delisting within one to two years of fiscal year-end t —and $Y_{it} = 0$ otherwise. Let $\mathbf{x}_{it}$ denote the vector of selected accounting and market predictors observed at fiscal year-end t , measured strictly before the default window. We model the conditional probability of default as

$$\Pr(Y_{it} = 1 \mid \mathbf{x}_{it}) = \frac{1}{1 + \exp[-(\alpha + \beta' \mathbf{x}_{it})]}. \quad (5)$$

The model is estimated by maximum likelihood on the firm-year panel. Each surviving firm-year contributes a non-default observation, and the last observed fiscal year of a distress-delisting firm contributes the single default observation for that firm. This specification is linear in the log-odds and therefore remains transparent: coefficient signs indicate whether a variable raises or lowers predicted default risk, holding the other predictors fixed. Because the same firm appears in multiple years, all standard errors and z -statistics are clustered by firm (`gvkey`) to account for serial correlation

across a firm’s repeated observations. We also emphasize out-of-sample predictive performance, calibration, and ranking metrics rather than relying only on coefficient-level significance.

3.2 XGBoost

The machine-learning model is XGBoost, a gradient-boosted tree ensemble (Chen and Guestrin, 2016). XGBoost builds an additive sequence of shallow decision trees, where each new tree is trained to improve the prediction errors of the existing ensemble. This structure allows the model to capture nonlinearities and interactions among predictors. For example, the effect of leverage may depend on profitability, stock-price performance, or equity volatility. Unlike the dynamic logit, XGBoost does not produce a single coefficient for each variable, so we evaluate it primarily through out-of-sample performance and feature-importance measures.

Two design choices are important. First, default is a rare event in the training sample, with an event rate of 1.404%. To address this class imbalance, we set `scale_pos_weight` equal to the ratio of non-default to default observations in the training sample, approximately 70.24. This gives more weight to default observations during training and improves rare-event ranking. However, class weighting can distort the level of raw predicted probabilities, so probability calibration is needed before interpreting model outputs as default probabilities.

Second, hyperparameters are selected by randomized search over 50 candidate configurations using a 5-fold time-series split within the training sample. The tuning objective is average precision, or AUC-PR, which is more informative than accuracy when defaults are rare. The selected configuration is shallow and regularized, with the full set of selected hyperparameters reported in Appendix Table A2.

3.3 Probability calibration

Because class weighting and boosted-tree training can make raw XGBoost probability levels unreliable, we recalibrate the fitted model using sigmoid calibration, also known as Platt scaling (Platt, 1999). The calibration step fits a monotone transformation from the XGBoost scores to observed default frequencies in the training data.¹ Ranking metrics such as AUC-ROC, AUC-PR, and KS are computed from model scores. Calibration ratios, confusion matrices, and expected-loss calculations based on XGBoost use calibrated predicted default probabilities.

3.4 Evaluation metrics

We evaluate the models along four dimensions. First, we measure *discrimination*, or ranking ability, using AUC-ROC, area under the precision-recall curve, and the Kolmogorov–Smirnov statistic.

¹The Platt calibrator is fit on the training sample to correct the probability inflation introduced by `scale_pos_weight`. It therefore recovers the probability level that prevailed in the training period, which is the relevant correction for a bias induced by class weighting. It does not protect against calibration drift when the mapping from firm fundamentals to realized default itself changes, as Section 5.1 shows it does around macro shocks, and no choice of calibration sample drawn from the training window would anticipate such a shift.

AUC-ROC summarizes how well the model ranks defaulting firms above non-defaulting firms. AUC-PR is especially relevant in this setting because defaults are rare (Davis and Goadrich, 2006). The KS statistic measures the maximum separation between the score distributions of defaulting and non-defaulting firms.

Second, we measure *goodness of fit* using McFadden-style pseudo- R^2 , computed from the binary cross-entropy log-likelihood of the fitted model relative to a null model. For the in-sample value, the null predicts the training-sample default rate. For the out-of-sample values, the null predicts the training base rate for every test observation—the only information available at the time of prediction. Anchoring the out-of-sample null to the training base rate, rather than to each window’s own realized default rate, keeps the statistic a genuine forecast benchmark and avoids letting future information enter the null. We report both in-sample and out-of-sample pseudo- R^2 .

Third, we assess *calibration* by comparing the average predicted probability of default with the realized default rate. A calibration ratio close to one means that the model’s average predicted default probability is close to the realized event rate. We compute this ratio for each test window and also within windows, separating years of acute macro-financial stress from ordinary years, because a window-level average can conceal offsetting periods of over- and underprediction.

Fourth, we compute a simple expected-loss measure,

$$EL = PD \times EAD \times LGD, \tag{6}$$

where PD is the predicted default probability used for the expected-loss calculation, EAD is proxied by short-term plus long-term debt, and LGD is fixed at 45%. We use 45% as an illustrative benchmark because the Basel Foundation IRB approach assigns a 45% LGD to senior unsecured corporate exposures (Basel Committee on Banking Supervision, 2019). This is a simplifying assumption: LGD is not estimated from the data, and the exercise is intended as a portfolio-level illustration rather than a full regulatory capital model.

3.5 Out-of-time design

All evaluation is out of time. The models are trained on observations through 2005 and then tested on three later, non-overlapping windows: 2006–2010, 2011–2015, and post-2015. This design is important because random cross-validation is not appropriate for the final evaluation of a default-prediction model. Random folds would mix future and past observations, allowing information from later macro-financial conditions to affect model assessment for earlier years. The out-of-time design instead asks whether a model trained on historical data can predict default risk in future periods.

4 Results

This section reports the empirical results. We first present the dynamic-logit benchmark, then summarize the XGBoost model-specific results, and finally compare the two approaches directly

using out-of-sample discrimination, pseudo- R^2 , and calibration.

4.1 Dynamic-logit benchmark

Table 3 reports the dynamic logit estimates using the training window. The model converges and has an in-sample pseudo- R^2 of 0.250. The coefficient signs are broadly consistent with standard economic intuition and with the default-prediction literature. Higher book leverage and market-scaled liabilities are associated with higher default risk, while stronger profitability, liquidity, firm size, stock price, and recent stock returns are generally associated with lower default risk.

The largest positive coefficient is on TLMTA, the ratio of total liabilities to market-valued assets. This is intuitive: firms whose liabilities are large relative to their market value are closer to distress. Book leverage also enters positively, suggesting that accounting leverage contains additional information beyond the market-valued liability measure. ROA, interest coverage, current ratio, CASHMTA, log assets, log price, and trailing 12-month return all enter with negative signs, while equity volatility enters positively. This is consistent with the idea that weak profitability, weaker liquidity and debt-service capacity, low valuation, poor recent stock performance, and high equity volatility are associated with greater default risk.

A few coefficients require more cautious interpretation. Cash/assets and NIMTA are not statistically significant in this specification—cash/assets enters with a small positive coefficient and NIMTA with a small negative one—likely because related information is already captured by CASHMTA, ROA, log price, and other distress measures. Market-to-book enters positively and significantly, which may reflect the behavior of high-valuation firms conditional on the rest of the risk controls rather than a simple unconditional relationship. Because logit coefficients are expressed in log-odds units, Appendix Table A3 reports marginal effects for one-standard-deviation changes in the predictors.

4.2 XGBoost results

The XGBoost model is trained on the same firm-year panel, selected predictor set, and outcome as the dynamic logit. Its out-of-sample performance is compared directly with the logit in Section 4.3. This subsection reports model-specific feature importance.

Figure 1 reports XGBoost feature importance based on total gain. Gain measures how much a feature improves the model’s objective when it is used in tree splits. It should therefore be interpreted as a measure of predictive usefulness, not as a coefficient or causal effect. The most important variables are log price, NIMTA, and ROA, followed by interest coverage and current ratio. This pattern is economically sensible: the boosted-tree model relies heavily on market valuation, profitability, liquidity, and debt-service capacity to rank default risk. Appendix Table A4 reports the corresponding numeric gain values.

The relatively lower gain assigned to some leverage variables does not imply that leverage is unimportant. In a tree model, importance can be spread across correlated predictors. Since TLMTA, book leverage, market-to-book, and other market-valued measures contain overlapping information

Table 3: Dynamic logit estimates using the training window

Variable	Coefficient	Std. err.
Constant	-3.545***	0.163
Book leverage	1.287***	0.157
Interest coverage	-0.001***	0.000
ROA	-0.655***	0.111
Current ratio	-0.152***	0.046
Cash/assets	0.251	0.336
Log assets	-0.314***	0.028
NIMTA	-0.045	0.178
TLMTA	1.905***	0.189
CASHMTA	-1.220**	0.492
Log price	-0.647***	0.067
Trailing 12-month return	-0.237***	0.065
Equity volatility	0.439***	0.076
Market-to-book	0.029***	0.005
Observations		103,154
Pseudo- R^2		0.250
Log-likelihood		-5709.0
Average firm-years per firm		7.78

Notes. The table reports coefficient estimates from the dynamic logit model estimated on the training sample, defined as firm-year observations through 2005. Standard errors, reported in the final column, are clustered by firm (**gvkey**); significance stars are based on these clustered standard errors: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The main comparison in the paper relies on out-of-sample performance.

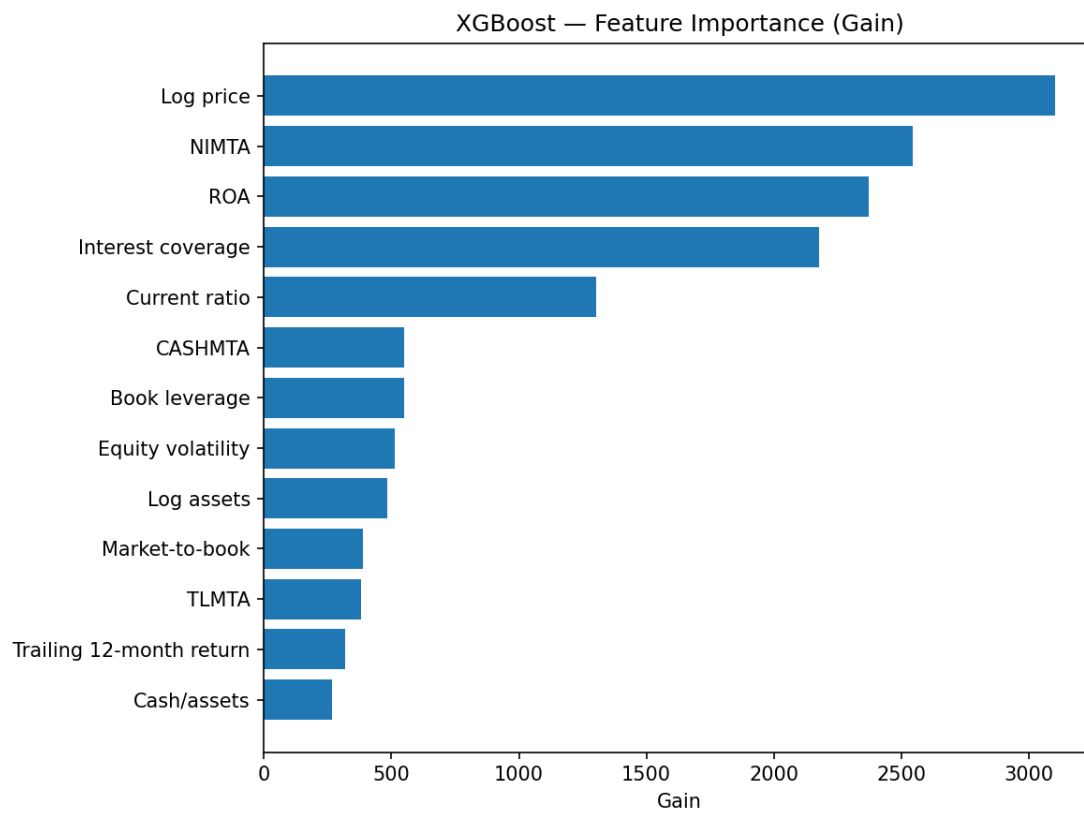


Figure 1: XGBoost feature importance based on gain

about firm distress, the model may use several related variables rather than assigning all predictive value to a single one. Feature importance should therefore be read as a descriptive summary of how the fitted model uses variables, not as a structural ranking of economic mechanisms.

Because the XGBoost model is trained with class weighting to address rare defaults, its raw predicted probabilities are not directly interpretable as calibrated default probabilities. We therefore apply sigmoid calibration to map the fitted XGBoost scores back to the realized default scale. This distinction is important: raw XGBoost scores are useful for ranking firms by risk, but calibrated probabilities should be used when comparing predicted default rates with realized default rates or when computing expected loss.

4.3 Model comparison

Discrimination and goodness of fit. Table 4 compares out-of-sample performance for the dynamic logit and XGBoost models across the three test windows. Both models discriminate well, with AUC-ROC values around 0.900–0.930. XGBoost posts higher AUC-ROC, AUC-PR, and KS than the dynamic logit in all three test windows. On AUC-ROC, however, the improvement is statistically significant only in 2011–2015 (DeLong test, $p = 0.001$); the 2006–2010 and post-2015 differences are not distinguishable from zero ($p = 0.611$ and $p = 0.180$). The most consistent and economically meaningful gains are in AUC-PR, the most relevant ranking metric in this rare-event setting: XGBoost improves AUC-PR substantially in 2011–2015 (0.097→0.148) and post-2015 (0.035→0.057), with a smaller edge in 2006–2010 (0.202→0.208). Because AUC-PR is bounded below by the default prevalence, these values should be read against their no-skill baselines of 0.011, 0.008, and 0.005, against which XGBoost runs roughly an order of magnitude above the random-classifier line in the rarest-event windows.

The pseudo- R^2 comparison is more even. Against a forecast null anchored to the training base rate, XGBoost improves out-of-sample pseudo- R^2 in all three windows, though only marginally in 2006–2010 (0.303 versus 0.307). The advantage is largest in the later windows, consistent with the AUC-PR pattern.

Figure 2 shows the ROC curves for both models in each test window. The figure confirms that both models rank defaulting firms well out of sample, with XGBoost providing a modest improvement in AUC-ROC that is statistically significant only in the 2011–2015 window.

Calibration. Discrimination is only part of the evaluation. A model can rank firms correctly while still producing probability levels that are too high or too low. Table 5 compares the average predicted probability of default with the realized default rate in each test window.

The dynamic logit underpredicts defaults in the first two test windows, with predicted-to-realized ratios of 0.648 and 0.766. It then overpredicts defaults after 2015, with a ratio of 1.997. XGBoost shows a similar pattern after sigmoid calibration. It is closer to the realized default rate in the first two windows, with ratios of 0.740 and 0.890, but it overpredicts more strongly after 2015, with a ratio of 2.073. The overprediction appears in both models, so it is not a problem unique to the

Table 4: Out-of-sample model comparison

Window	Model	AUC-ROC	AUC-PR	KS	Pseudo- R^2	PR base.
2006–2010	Dynamic logit	0.924	0.202	0.710	0.303	0.011
2006–2010	XGBoost	0.927	0.208	0.733	0.307	
2011–2015	Dynamic logit	0.902	0.097	0.698	0.252	0.008
2011–2015	XGBoost	0.930[†]	0.148	0.730	0.306	
Post-2015	Dynamic logit	0.906	0.035	0.720	0.203	0.005
Post-2015	XGBoost	0.916	0.057	0.723	0.276	

Notes. Bold entries indicate the better value within each test window and metric. AUC-PR is area under the precision-recall curve; “PR base.” is the no-skill precision-recall baseline, equal to the realized default prevalence in that window. KS is the Kolmogorov–Smirnov statistic. Pseudo- R^2 is computed relative to a null model that predicts the training-sample base rate, so values are comparable across models within a window. [†] The AUC-ROC difference between XGBoost and the logit is significant at the 1% level by a DeLong paired test; window p-values are 0.611 (2006–2010), 0.001 (2011–2015), and 0.180 (post-2015).

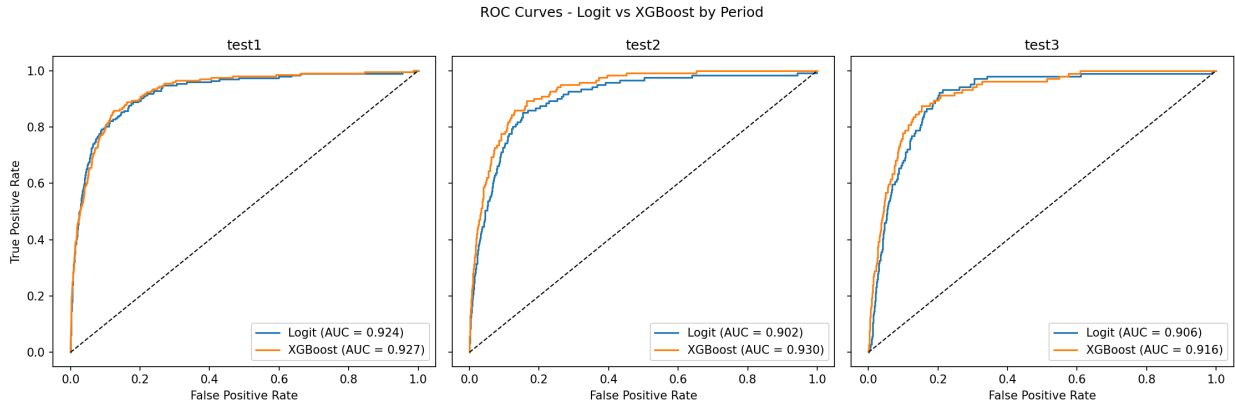


Figure 2: ROC curves by test window: dynamic logit versus XGBoost

machine-learning model. It is also not a property of the post-2015 period as a whole.

Table 5: Calibration: average predicted probability versus realized default rate

Window	Model	Avg. predicted PD	Realized rate	Ratio
2006–2010	Dynamic logit	0.007	0.011	0.648
2006–2010	XGBoost, calibrated	0.008	0.011	0.740
2011–2015	Dynamic logit	0.006	0.008	0.766
2011–2015	XGBoost, calibrated	0.007	0.008	0.890
Post-2015	Dynamic logit	0.010	0.005	1.997
Post-2015	XGBoost, calibrated	0.010	0.005	2.073

Notes. The ratio is the average predicted probability of default divided by the realized default rate in the same test window. A ratio above one indicates overprediction; a ratio below one indicates underprediction. Ratios are computed before rounding the displayed probabilities. The dynamic logit is the raw MLE fit with no post-hoc calibration; XGBoost probabilities are Platt-calibrated using a sigmoid transformation fit on the training sample.

Table 6 splits each test window into years of acute macro-financial stress and ordinary years. Two episodes account for most of the calibration failure. In 2007–2008 both models underpredict by roughly half, with ratios of 0.51 and 0.56, while the other three years of the 2006–2010 window sit close to one (0.96 and 1.13). In 2020–2022 the error reverses and is far larger: average predicted probabilities near 0.011 against a realized distress-delisting rate of 0.0006, a ratio close to 17 for both models. The 2016–2019 years are again well calibrated (1.07 and 1.15), as is the whole of 2011–2015 (0.77 and 0.89), the one window that contains no stress episode. The window-level ratios in Table 5 are averages over subsamples that behave very differently, rather than descriptions of a uniform regime.

The 2020–2022 ratio deserves a caveat. It rests on roughly five to six realized default events, because the firm-year base in those years is small and the realized rate is close to zero. The direction and the rough scale of the miscalibration are clear, but the point value is not precise, and we use it as evidence that the models fail badly in that episode rather than as an estimate of how badly.

The stress years do not, however, account for the discrimination pattern. Recomputing the DeLong test on the ordinary years leaves the 2006–2010 comparison essentially unchanged ($p = 0.631$, against $p = 0.611$ for the full window), and moves the post-2015 comparison only to $p = 0.113$, still short of conventional significance. The AUC-ROC advantage is modest and statistically inconclusive outside 2011–2015 whether or not stress years are included, so it should not be attributed to stress-year noise. AUC-PR is the exception. Dropping 2020–2022 raises it from 0.035 to 0.066 for the logit and from 0.057 to 0.094 for XGBoost, since that subsample is an extreme rare-event slice that mechanically depresses precision-recall performance for both models.

Overall, machine learning predicts corporate default somewhat better than the traditional dynamic logit benchmark on this panel. XGBoost has higher AUC-ROC, AUC-PR, and KS in every test window and higher out-of-sample pseudo- R^2 in all three, but the AUC-ROC edge is

Table 6: Calibration within test windows: stress years versus ordinary years

Window and years	Model	Avg. predicted PD	Realized rate	Ratio
<i>2006–2010</i>				
Stress (2007–2008)	Dynamic logit	0.0098	0.0193	0.51
	XGBoost, calibrated	0.0109	0.0193	0.56
Ordinary (2006, 2009–2010)	Dynamic logit	0.0057	0.0060	0.96
	XGBoost, calibrated	0.0067	0.0060	1.13
<i>2011–2015</i>				
All years (no stress episode)	Dynamic logit	0.0060	0.0078	0.77
	XGBoost, calibrated	0.0070	0.0078	0.89
<i>Post-2015</i>				
Stress (2020–2022)	Dynamic logit	0.0107	0.0006	17.1
	XGBoost, calibrated	0.0107	0.0006	17.1
Ordinary (2016–2019)	Dynamic logit	0.0087	0.0082	1.07
	XGBoost, calibrated	0.0094	0.0082	1.15

Notes. The table reports the same calibration ratio as Table 5, computed separately for years of acute macro-financial stress and for the ordinary years within each test window. Stress years are the global financial crisis (2007–2008) and the COVID-19 period and its immediate aftermath (2020–2022); the 2011–2015 window contains no such year, so it is reported as a single block and its entries match Table 5. Probabilities are shown to four decimal places because the realized rate in 2020–2022 is close to zero. That row rests on roughly five to six realized default events, so its ratio should be read as directional rather than as a precise point estimate. Models are defined as in Table 5.

statistically significant only in 2011–2015, and most of the point-estimate gaps are small. The clearest improvement is in the rare-event metric, AUC-PR. Calibration behaves differently from ranking. Neither model’s probability level is uniformly right or uniformly wrong: both track realized default rates closely in ordinary years and fail in opposite directions during the two stress episodes. For credit-risk applications this separates the two uses of these models. Ranking is stable enough to screen on across every window tested here, while the probability levels that feed expected-loss calculations hold only in the conditions they were fit for. Section 5 takes up why the two failures run in opposite directions.

5 Discussion

5.1 Calibration around macro shocks

The main results show a clear distinction between ranking performance and probability calibration, and the second of these fails in a specific and limited way. Both models rank firms well in every test window, as shown by the AUC-ROC and KS statistics in Table 4. Their probability levels are also close to realized default rates in most years. Table 6 locates nearly all of the calibration failure in two episodes, 2007–2008 and 2020–2022, and the two failures run in opposite directions. Across the twelve ordinary years, calibration ratios fall between 0.77 and 1.15 for both models. The window-level ratios of 1.997 and 2.073 reported for the post-2015 period in Table 5 are averages over four well-calibrated years and three badly miscalibrated ones, not a description of a period

that drifted.

Figure 3 shows the same pattern year by year. The predicted and realized series track each other for most of the sample and separate sharply at the two episodes, with predictions falling below realized rates around the financial crisis and rising well above them in the pandemic years.

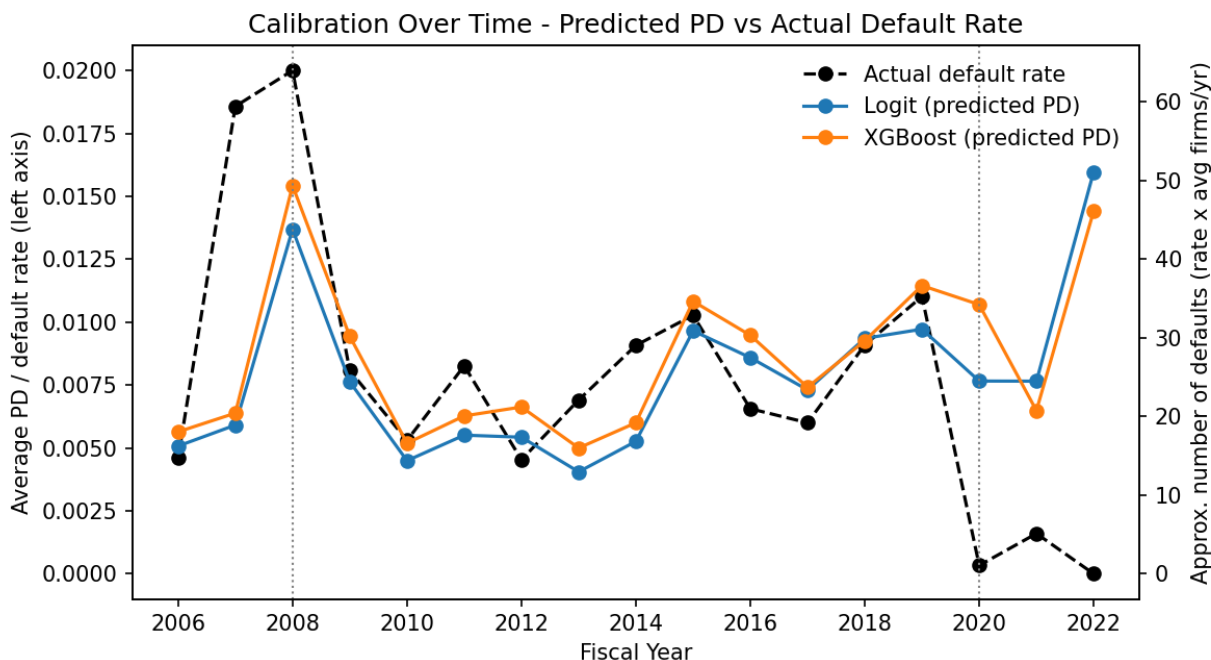


Figure 3: Calibration over time: predicted probabilities versus realized default rates. The dynamic logit is the raw MLE fit with no post-hoc calibration; XGBoost probabilities are Platt-calibrated using a sigmoid transformation fit on the training sample.

The 2007–2008 failure is one of speed. Both models are trained on data through 2005, a period that contains recessions but no systemic financial crisis, and they learn the historical mapping from firm-level distress signals to realized default. When credit conditions tightened in 2007 and 2008, realized distress arrived faster than that mapping implied, and firms defaulted at roughly twice the rate the fitted probabilities anticipated. The ordering of firms by risk was still right; the level was too low. This is what one would expect of a model asked to extrapolate past the most severe episode in its training sample.

The 2020–2022 failure runs the other way and has a different source. Much of the post-2015 period was more refinancing-friendly than the training period, with relatively low interest rates and policy rates that returned to near zero during the pandemic. Firms with high leverage, weak profitability, or low stock prices continued to look risky on historical distress signals while remaining able to refinance debt, extend maturities, or otherwise avoid an immediate distress-related delisting. During the pandemic this was reinforced by extraordinary support for corporate credit markets. The Federal Reserve created corporate credit facilities in March 2020 to support bond issuance and secondary-market liquidity, and firms raised unusually large amounts of debt after these facilities

were announced ([Federal Reserve Board, 2020a,b](#); [New York Fed, 2020](#)). Policy loosened the link between fundamentals and realized default, and the models, which observe only fundamentals, continued to predict from the historical relationship.

Read together, the two episodes describe one limitation rather than two unrelated ones. A model fitted to an ordinary-times mapping from fundamentals to default underpredicts when a shock arrives faster than anything in its estimation sample, and overpredicts when policy intervention severs the link between fundamentals and realized default. Both are changes in the mapping itself rather than deterioration in the model’s ability to separate risky firms from safe ones, which is why discrimination survives both episodes while the probability level fails in both. The direction of the error is not a fixed property of the model. It depends on which kind of regime change is underway, so its sign cannot be signed in advance, and that is harder to manage in practice than a bias of known direction.

The outcome definition contributes to the 2020–2022 gap. Our default measure captures CRSP distress-related delistings, so distress that resolved through refinancing, restructuring, acquisition, a going-private transaction, or a bankruptcy delayed beyond the two-year window does not register as an event. Part of what appears as overprediction may be economic fragility the models identified correctly but that did not surface as a delisting in those years.

The gap between ranking and level is what connects this to stress testing. Training a model before a crisis and evaluating it through one is close in spirit to a severely adverse supervisory scenario, and on the ranking dimension both models pass: AUC-ROC holds near 0.91–0.93 through 2007–2008, so the ordering of firms by risk comes through the crisis intact. The calibrated probability level does not, understating realized defaults by about half. A model that ranks well through a crisis is therefore not automatically one a bank can rely on under a severely adverse scenario, since capital and provisioning depend on the level rather than the ordering. Judged on discrimination alone, the failure documented here would not have been visible at all.

Supervisory and accounting practice already assumes as much, and our results are an empirical instance of why. Through-the-cycle rating systems exist in part to avoid the point-in-time swings visible in Table 6; CCAR and DFAST require banks to test against several prescribed macro scenarios rather than a single historical average; and IFRS 9 and CECL require scenario-weighted, forward-looking provisions rather than one expected path ([Financial Accounting Standards Board, 2016](#)). The revised interagency guidance on model risk management, which replaced SR 11-7 in April 2026, asks for ongoing monitoring and outcomes analysis over a model’s life ([Board of Governors et al., 2026](#)), and the split in Table 6 is a small example of what such analysis is for: a window-level ratio near two, which reads as gradual drift, resolves into four accurate years and three badly inaccurate ones. The expected-loss calculation in Section 5.2 is in the same spirit as CECL provisioning, though it applies a one- to two-year forward PD rather than the lifetime loss estimate that CECL requires. Regime dependence is well understood in principle; what this exercise adds is a measured instance of it in an otherwise standard model on standard data.

5.2 Expected-loss implications

The expected-loss framework is useful for organizing the economic meaning of the predictions:

$$EL = PD \times EAD \times LGD.$$

In this paper, the models provide the PD component, exposure at default is proxied by total debt, and LGD is fixed at 45%. We use 45% as an illustrative benchmark because the Basel Foundation IRB approach assigns a 45% LGD to senior unsecured corporate exposures (Basel Committee on Banking Supervision, 2019). This is a simplifying assumption: LGD is not estimated from the data, and the exercise is intended as a portfolio-level illustration rather than a full regulatory capital model. We report average annual EAD and EL within each evaluation window, aggregating firm-level exposures within each fiscal year before averaging across the years in the window, so the figures reflect a point-in-time portfolio snapshot rather than a cumulative multi-year sum. The calculation is still informative because it shows how differences in calibrated predicted default probabilities translate into economically meaningful differences in expected losses.

Table 7 reports expected losses by test window for both models. XGBoost produces higher expected losses than the dynamic logit in every test window, and by a wider margin than the average predicted probabilities suggest. In the post-2015 window the two average PDs differ by about four percent of each other, while expected losses differ by a factor of two. The reason is that expected loss weights each firm’s predicted probability by that firm’s own debt, and XGBoost assigns relatively higher probabilities to firms carrying large balances. The EL/EAD column shows this directly: the implied loss rate under XGBoost is roughly twice the logit’s in each window (0.09% versus 0.17%, 0.08% versus 0.15%, and 0.07% versus 0.15%). The difference is economically meaningful. In the post-2015 window, for example, the dynamic logit implies average annual expected losses of about \$6.2 billion, while calibrated XGBoost probabilities imply about \$13.2 billion.

The post-2015 figures inherit the calibration problem identified in Section 5.1. Average predicted probabilities in that window pool four well-calibrated years with three in which predictions exceed realized defaults by more than an order of magnitude, so the reported level of expected loss is higher than the ordinary-year probabilities alone would justify. The comparison between the two models is less affected: their predicted probabilities are nearly identical in 2020–2022 and differ mainly in the ordinary years, so the stress subsample dilutes the gap rather than creating it. A reader using these figures as portfolio estimates rather than as a model comparison should treat the post-2015 row accordingly.

Figure 4 shows the same expected-loss comparison over time. Expected losses rise in years when predicted default probabilities and debt exposure are high. The XGBoost series is generally above the dynamic-logit series, reflecting the higher calibrated PDs used by XGBoost in several years.

The main lesson is that ranking performance alone is not enough for expected-loss applications. XGBoost improves the rare-event ranking metric (AUC-PR) in the later windows and out-of-sample pseudo- R^2 in all three, although its AUC-ROC advantage is statistically significant only in 2011–2015.

Table 7: Expected-loss implications

Window	EAD (\$B)	Dynamic logit			XGBoost		
		Avg. PD	EL (\$B)	EL/EAD	Avg. PD	EL (\$B)	EL/EAD
2006–2010	4,065.5	0.74%	3.7	0.09%	0.84%	6.8	0.17%
2011–2015	5,549.2	0.60%	4.4	0.08%	0.70%	8.3	0.15%
Post-2015	8,659.4	0.96%	6.2	0.07%	1.00%	13.2	0.15%

Notes. *EAD* and *EL* are reported as annual averages within each window: expected loss is computed firm by firm as $PD \times EAD \times LGD$, summed within each fiscal year, and then averaged across the years in the window, so the figures represent a point-in-time portfolio rather than a cumulative multi-year total. Average PD is the unweighted mean predicted probability across all firm-years in the window and is reported as a descriptive summary of the probability level. Because *EL* weights each firm's probability by that firm's own exposure while average PD does not, the product of the reported average PD, *EAD*, and *LGD* does not reproduce the reported *EL*; the two columns describe different quantities. *EAD* is proxied by short-term plus long-term debt. *LGD* is fixed at 45%, following the Basel Foundation IRB benchmark for senior unsecured corporate exposures (Basel Committee on Banking Supervision, 2019). Expected losses use calibrated predicted default probabilities for XGBoost. Dollar amounts are rounded to the nearest \$0.1 billion.

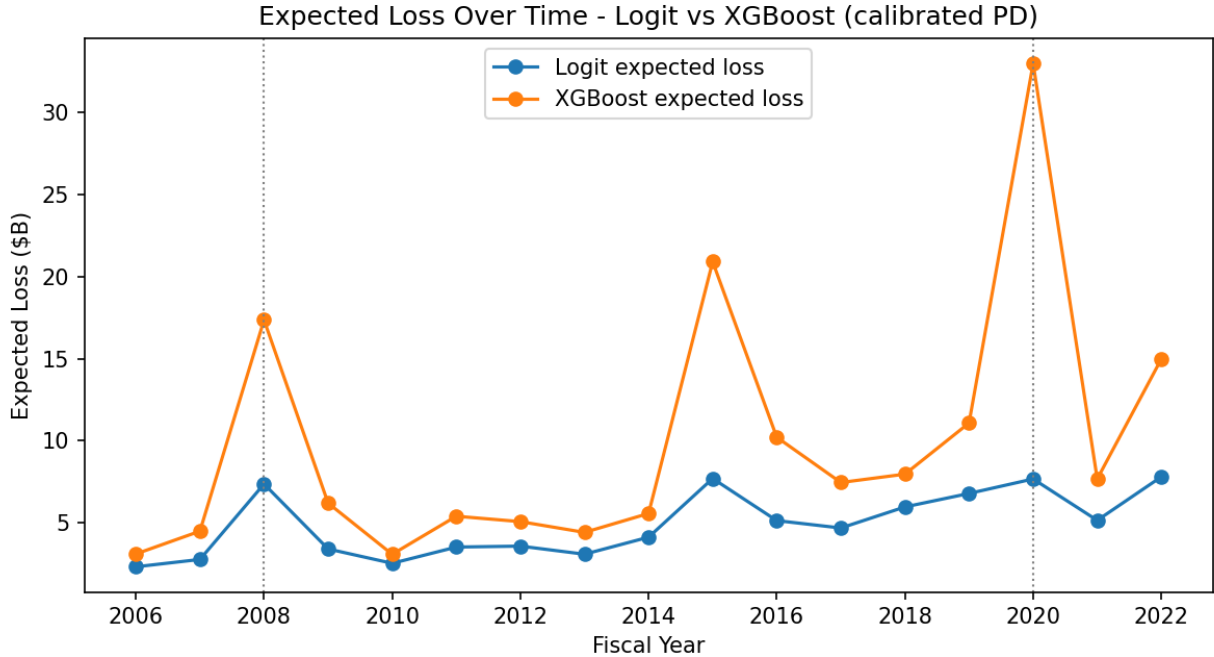


Figure 4: Expected loss over time: dynamic logit versus XGBoost

Expected loss depends on calibrated probabilities rather than on rankings alone. A calibration step is therefore necessary before such a model is used for provisioning, portfolio monitoring, or stress testing, though Section 5.1 shows it is not sufficient. The probabilities used here were Platt-calibrated on the training period and still failed in both directions once the mapping from fundamentals to realized default changed. What the calibration step cannot supply is outcome monitoring fine enough to catch that failure while it is happening.

The broader implication is that machine learning is most valuable here as a screening and ranking tool. It improves the ordering of firms by default risk, especially in a rare-event setting, but it does not remove the need for probability calibration, period-by-period validation, and judgment about the default environment.

6 Conclusion

This paper compares a traditional dynamic logit default model with an XGBoost classifier using the same firm-year panel, selected predictor set, and default outcome. The dynamic logit follows the logic of the standard default-prediction literature and remains a strong, transparent benchmark. XGBoost adds flexibility by allowing nonlinearities and interactions among the same accounting and market variables.

The results show that machine learning predicts corporate default somewhat better, but not dramatically better. XGBoost achieves higher AUC-ROC, precision-recall AUC, and KS in every test window and higher out-of-sample pseudo- R^2 (against a forecast null) in all three, but the AUC-ROC edge is statistically significant only in 2011–2015 (DeLong test, $p = 0.001$), and most point-estimate gaps are small. Excluding the stress years does not change this: the comparison remains statistically inconclusive in the other two windows either way. The improvement is most useful and most consistent for rare-event ranking, where precision-recall performance is especially informative. At the same time, the dynamic logit remains highly competitive, transparent, and easier to interpret.

The main weakness of both models is calibration, and it is a weakness of a particular shape. Probability levels sit close to realized default rates in ordinary years and fail in both directions during macro shocks. The models understate defaults by roughly half in 2007–2008 and overstate them by about an order of magnitude in 2020–2022, the second of these resting on few enough events that its magnitude is indicative rather than precise. Ranking survives both episodes. What separates the two failures is not the model but the regime: a shock that outruns the training data pushes predicted probabilities below realized defaults, while policy support that decouples defaults from fundamentals pushes them far above. For practical credit-risk applications the conclusion is therefore balanced. Flexible machine-learning models can improve default screening, but expected-loss applications need probabilities validated against the macro environment they are used in, since a calibration ratio averaged over a multi-year window can hide exactly the periods when the number matters most.

A Additional Tables and Figures

This appendix reports additional descriptive and diagnostic tables and figures that support the main analysis but are not central to the paper’s main results.

Table A1: Default events and rates by period

Period	Firm-years	Defaults	Rate (%)
Train (≤ 2005)	103,154	1,448	1.404
2006–2010	17,298	197	1.139
2011–2015	15,475	121	0.782
Post-2015	21,629	104	0.481
Full panel	157,556	1,870	1.187

Notes. Default events are proxied by CRSP distress-related delisting codes: **BKPY**, **INSC**, and **DELQ**. Each event is assigned to the firm’s last complete fiscal year in the cleaned annual panel, retained only when that fiscal year precedes the delisting year by one or two years.

Table A2: XGBoost hyperparameters selected by time-series split

Hyperparameter	Selected value
<code>max_depth</code>	2
<code>n_estimators</code>	300
<code>learning_rate</code>	0.050
<code>subsample</code>	0.600
<code>colsample_bytree</code>	0.700
<code>min_child_weight</code>	10
<code>reg_lambda</code>	10
<code>scale_pos_weight</code>	70.240
Best cross-validated AUC-PR	0.169

Notes. Hyperparameters are selected by randomized search over 50 candidate configurations using a 5-fold time-series split within the training sample. The tuning objective is average precision, equivalent to AUC-PR. The class-weight parameter `scale_pos_weight` is set to the non-default/default ratio in the training sample.

Table A3: Dynamic-logit marginal effects for one-standard-deviation changes

Variable	Train	2006–2010	2011–2015	Post-2015
Book leverage	0.337	0.187	0.154	0.256
Interest coverage	-0.170	-0.170	-0.142	-0.240
ROA	-0.236	-0.124	-0.092	-0.190
Current ratio	-0.555	-0.277	-0.242	-0.421
Cash/assets	0.062	0.039	0.031	0.051
Log assets	-0.850	-0.444	-0.364	-0.559
NIMTA	-0.010	-0.005	-0.004	-0.008
TLMTA	0.616	0.323	0.259	0.394
CASHMTA	-0.230	-0.150	-0.113	-0.215
Log price	-0.612	-0.313	-0.245	-0.378
Trailing 12-month return	-0.217	-0.113	-0.065	-0.123
Equity volatility	0.184	0.089	0.056	0.126
Market-to-book	0.142	0.074	0.085	0.191

Notes. Entries report percentage-point changes in predicted default probability associated with a one-standard-deviation increase in each predictor, computed within each period. For example, 0.328 corresponds to approximately 0.328 percentage points. Figure A1 plots the training-window effects in probability units.

Table A4: XGBoost feature importance based on gain

Feature	Gain
Log price	3104.69
NIMTA	2545.35
ROA	2373.55
Interest coverage	2178.99
Current ratio	1303.78
CASHMTA	551.16
Book leverage	547.95
Equity volatility	514.19
Log assets	485.33
Market-to-book	389.61
TLMTA	382.52
Trailing 12-month return	318.81
Cash/assets	266.06

Notes. Gain summarizes the improvement in the XGBoost objective from splits using a given feature. It measures predictive usefulness within the fitted model and does not indicate the sign of the relationship between the feature and default risk.

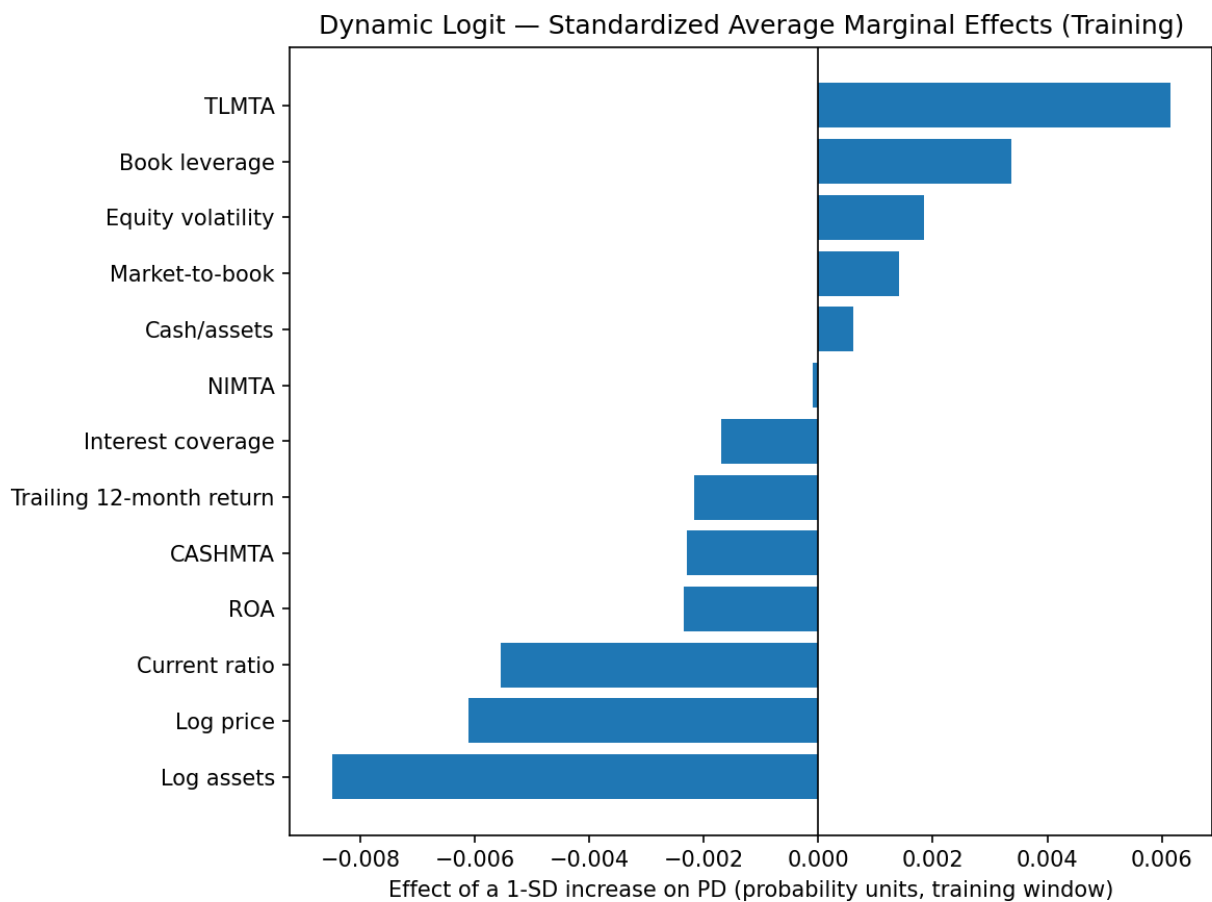


Figure A1: Dynamic-logit standardized average marginal effects (training window). Effects are shown in probability units; Table A3 reports the same quantity in percentage points.

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